

Tools interlude

R vs Python, without the holy war

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R vs Python: what to expect today

Most of you code in **Python**.

Some key finance / QuantLet pipelines (incl. Crypto Zombies) are in **R**.

Today everything runs in Python. If you work in R, the design carries over line by line; only the package names change.

What actually matters

The scarce skill is **research design**:

label · information set · OOT validation · economic metric · leakage

Language is secondary — as long as you can reproduce the design.

Tooling map: today vs the papers vs your thesis

Task	Today	In the papers	In your own work
Penalised regression	sklearn	glmnet (R), sklearn	sklearn
Random forest, boosting	sklearn	H2O in R (Krauss et al.)	lightgbm , xgboost
Neural nets	sklearn MLP	TensorFlow / Keras (GKX, CPZ)	torch
Economic structure in the loss	numpy sketch (04)	TensorFlow (Chen–Pelger–Zhu)	torch
Variable importance	permutation	SHAP, gradient-based	shap
Inference on returns	numpy	Newey–West, bootstrap	statsmodels , arch

Small stack on purpose: it installs in minutes and every live run ends in under two. **No conclusion today depends on the framework.**

Environment hygiene

Our `requirements.txt` did not list `scipy`. Two notebooks import it; it arrived only as a **scikit-learn dependency**. Everything ran, so nothing looked wrong.

- An implicit dependency is the most common reason a replication package "works on my machine".
- The test that finds it: **fresh environment, install from the file, run everything.**
- Ship the environment file with the code, and say which Python you used.

We come back to this in Block D, where we ask what makes a result replicable.

Practical split

Lean Python	Lean R
sklearn / PyTorch workflows	glm / glmnet / ranger idioms
Industry & CS collaborators	Many academic finance scripts
APIs, dashboards, DL	Tidy diagnostics & CRAN vignettes
Today's live notebooks	The QuantLet package behind Block E

Where the files live

```
notebooks/*.ipynb    ← everything we run today, in Python
data/                ← the data, with precomputed results in data/cache/
scripts/             ← the slow steps, precomputed
```

The Crypto Zombies replication package (Block E) is **R**, from QuantLet; today's classroom version of it is **Python**.

Classroom rule

1. Live coding today is **Python only**.
2. If you work in R, the design translates line by line: label, split, metric.
3. Grade / discuss **design**, not package taste.

One-liner

Use Python for the stack you ship; use R when the literature / QuantLet already did; always defend the **validation and the loss**, not the logo on the IDE.